

CONTACT INFORMATION Department of Mathematics website: <https://nick-lohr.github.io/>
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West Lafayette, IN 47907-2067 USA

APPOINTMENTS **Golomb Visiting Assistant Professor** Fall 2025 - present
Purdue University

EDUCATION **Northwestern University** September 2019 - June 2025
Ph.D. in Mathematics,
Thesis advisors: Jared Wunsch and Steven Zelditch

University of Notre Dame August 2015 - May 2019
B.S. in Mathematics with minor in Actuarial Science, *magna cum laude*
Senior thesis advisor: Jeffrey Diller

RESEARCH INTERESTS Analysis of PDE, semiclassical analysis, spectral and scattering theory, mathematical physics and Wigner distributions, harmonic analysis, probability, quantum dynamics.

RESEARCH **Papers**

- *The asymptotic structure of forward scattering* (with Ethan Sussman, Izak Oltman, Joey Zou), submitted.
- *Semiclassical measures through Coulomb collisions*, submitted.
- *On the intertwining map between Coulomb and hyperbolic scattering*, Analysis & PDE **19**(2026), no. 6, 1225-1250.
- *Semiclassical measures of eigenfunctions of the attractive Coulomb operator*, Annales Henri Poincaré **27**(2026), 2743-2768
- *Scaling asymptotics of Wigner distributions of harmonic oscillator orbital coherent states*, Communications in Partial Differential Equations **48**(2023), 415-439.

Talks

- Great Lakes Mathematical Physics Meeting Dayton, Summer 2026
- AMS 2026 Spring Eastern Sectional Meeting Boston College, Spring 2026
- Northwestern Analysis Seminar Northwestern, Winter 2026
- Spectral and Scattering Theory Seminar Purdue, Spring 2025
- Northwestern Informal Probability Seminar Northwestern, Fall 2024
- Northwestern Undergraduate Math Society Northwestern, Winter 2024
- RTG Dynamics REU Northwestern, Summer 2023
- 87th Midwest PDE Seminar Notre Dame, Spring 2023
- Microlocal Seminar Northwestern, Spring 2023
- Graduate Student Seminar Northwestern, Spring and Fall 2021, Fall 2022

TEACHING **Purdue Teaching** Fall 2025 - present
West Lafayette, IN

- MA266 Differential Equations (two sections: 083 and 072) Fall 2026
- MA265 Linear Algebra (two sections: 225 and 214) Spring 2026
- MA265 Linear Algebra (two sections: 501 and 206) Fall 2025

Northwestern Teaching Assistant Fall 2019 - Spring 2024
Evanston, IL

- Assisted teaching and grading for the following classes

- Single-Variable Calculus with Precalculus (MATH 218-3) Spring 2024
 - with Sonja Mapes
- Probability and Stochastic Processes (MATH 310-3)
 - with Ursula Porod
- Single-Variable Calculus with Precalculus (MATH 218-2) Winter 2024
 - with Sonja Mapes
- Probability and Stochastic Processes (MATH 310-2)
 - with Keith Burns
- Single-Variable Calculus with Precalculus* (MATH 218-1) Fall 2023
 - with Sonja Mapes
- Single-Variable Calculus with Precalculus (MATH 218-1) Fall 2022
 - with Sonja Mapes
- Probability Theory and Stochastic Analysis (MATH 450-1)
 - with Antonio Auffinger
- MENU Applied Analysis (MATH 360-2) Winter 2022
 - with Eugene Kushnirsky
- Complex Analysis and Group Theory for ISP (MATH 382-0)
 - with Ezra Getzler
- MENU Applied Analysis (MATH 360-1) Fall 2021
 - with Eugene Kushnirsky
- Fourier Analysis and Boundary Value Problems for ISP (MATH 381-0)
 - with Santiago Cañez
- Single-Variable Calculus with Precalculus (MATH 218-2) Spring 2021
 - with Sean McAfee
- MENU Probability and Stochastic Processes (MATH 311-3)
 - with Ursula Porod
- Probability Theory and Stochastic Analysis (MATH 450-3)
 - with Antonio Auffinger
- Sequences and Series (MATH 226-0) Fall 2020
 - with Eric Zaslów
- Elementary Differential Equations (MATH 250-0)
 - with Eric Chang
- Probability and Stochastic Processes[†] (MATH 310-2) Winter 2020
 - with Keith Burns

COMMUNITY
SERVICE

RTG Dynamics REU Advisor

Summer 2023

Evanston, IL

- Advised undergraduates biweekly in undertaking and presenting research in Hamiltonian dynamics

Northwestern Directed Reading Program

Spring 2021 - Winter 2023

Evanston, IL

- Advised undergraduates in undertaking and presenting independent study projects in mathematics
 - Levi Hoogendoorn - Grace-Heawood theorems Winter 2023
 - Claire Paré - Isoperimetric and isodiametric inequalities
 - Colin Brennan - Lagrange inversion theorem with applications in analysis, algebra, and topology Winter 2022
 - Kole Butterer - Brownian motion and the Dirichlet problem with SageMath visualization Spring 2021

*two sections

[†]only graded

Northwestern Causeway Certificate Program Mentor Summer 2023-Summer 2025

Evanston, IL

- Advised the following post-baccalaureate students in preparing for graduate school
 - Xzavier Herbert

Northwestern Climate Committee Member

2022-2023

Evanston, IL

- Met quarterly to discuss how to improve graduate students well-being with faculty members

Head Juror of First Degree Murder Trial

Winter 2022

Chicago, IL

- Served jury duty for Cook County and was voted to be the jury's foreperson

AWARDS AND
GRANTS

Nominated by Math Department for the University Teaching Award	2024
NSF RTG Dynamics: Classical, Modern, and Quantum Recipient	2022-2023
Department Graduate Student Teaching Assistant Award	2021-2022
NSF RTG: Analysis on Manifolds Recipient	2020-2022
GE Prize for Excellence in Mathematics - Honors	2019
College of Science Summer Undergraduate Research Fellowship	2018
Riverbend Math Center Award for Excellence in Quantitative Analysis	2015
Bayer Physical Science Award for Excellence in Physical Science	2015
Hoosier Science-Engineering Fair Award	2014
Mu Alpha Theta Science Fair Award	2014
University of Notre Dame Mathematics Department Award	2014

PROGRAMMING
SKILLS

LaTeX	Proficient	■■■□
Mathematica	Proficient	■■■□
Python/SageMath	Intermediate	■■□□
Julia	Intermediate	■■□□
C++	Elementary	■□□□
Emacs/LISP	Elementary	■□□□
HTML/Javascript	Elementary	■□□□